Michael C. Anoruo
Pittsburgh, PA | manoruo@andrew.cmu.edu | manoruo.github.io

Education

Carnegie Mellon University (CMU) Pittsburgh, PA
Ph.D., Robotics – Advisor: Wennie Tabib (RISLab) *Fall 2024 – Present*

University of Maryland, Baltimore County (UMBC) Baltimore, MD
B.S., Computer Science (AI/ML track); GPA: 3.93/4.0 *Fall 2020 – December 2023* **Relevant Coursework:** Machine Learning, Intro to AI, Algorithm Design & Analysis, Computer Vision (Grad), Data Science (Grad), Natural Language Processing (Grad), Operating Systems, Linear Algebra, Software Engineering I

Honors / Awards

GEM Fellow	Fall 2024
Summa Cum Laude	Winter 2023
Google CS Research Mentorship Program Scholar	Spring 2022
MIT Quantitative Methods Workshop	Winter 2022
NIDA EDUCATE Scholar	Fall 2021
Meyerhoff Scholar	Fall 2020
National Football Foundation Scholar Athlete	Spring 2020

Skills

Programming Languages: Python, C++, Lua, JavaScript, C, C#, R
ML / RL: PyTorch, TensorFlow, Hugging Face, OpenAI Gym, PPO, SAC
Robotics: ROS, Quadrotor Modeling & Control, Differentiable Simulation, 3D Reconstruction
Systems: Linux, macOS, Windows, HPC Clusters (SLURM)

Relevant Experience

Carnegie Mellon University (RISLab): Graduate Research Assistant Pittsburgh, PA
Fall 2024 – Present
- Investigating analytical policy gradients for reinforcement learning-based control of quadrotor drones across various tasks.
- Developed a differentiable simulator for training RL policies with analytical gradients.
- Co-authored paper submitted to CoRL 2025 (under review).

Carnegie Mellon University: Teaching Assistant, Mobile Robots Pittsburgh, PA

- Served as TA for Mobile Robots on two occasions.

Johns Hopkins University APL: Research Intern
2024
- TODO: Add description

UMBC (DPrime.ai Lab): Data Scientist Intern Baltimore, MD
12/2022 – 5/2024
- Assisted in developing machine learning algorithms to analyze eye-tracking data and identify patterns linked to cognitive events.
- Developed software for real-time visualization of data collected from various biometric sensors.

Apple Inc. (Apple Service Engineering): Data Engineer Intern Seattle, WA
5/2023 – 8/2023
- Developed code that worked with large volumes of video content, utilizing computer vision and machine learning techniques to automate video analysis processes.
- Created various analysis tools that helped data engineers optimize their viewership reporting algorithms.

MIT (Space Enabled Media Lab): Zero Robotics Research Intern Cambridge, MA

6/2022 – 8/2022

- Optimized PID controller parameters for the MIT Astrobee robot simulator.
- Developed Python scripts to automate data collection and analysis for the Zero Robotics project.
- Worked in small team of 6, alongside NASA astronauts, to host the Zero Robotics 2022 technical competition.

UPENN (Scalable Autonomous Robots Lab): Robotics Research Intern Philadelphia, PA

6/2021 – 8/2021

- Created a PCB with KiCad to translate RS232 to USB for data transmission.
- Developed simulation framework in Python to simulate Autonomous Surface Vehicles (ASV) experiments.
- Designed waypoint algorithm to generate strategic path for lab's ASV to survey Schuylkill River.

UMBC (Ebiquity Lab): Cybersecurity Research Intern Baltimore, MD

9/2020 – 5/2022

- Designed reinforcement learning algorithm to find vulnerabilities in malware classifiers by modifying malicious files to avoid detection.
- Annotated cybersecurity related texts to train machine learning algorithm to detect and classify such phrases.
- Designed algorithm to identify vulnerabilities given a particular operating system.

Publications

Piplai, A., **Anoruo, M.**, Fasaye, K., Joshi, A., Finin, T., & Ridley, A. (2022). Knowledge guided two-player reinforcement learning for cyber attacks and defenses. In *2022 21st IEEE International Conference on Machine Learning and Applications (ICMLA)* (pp. 1342–1349). IEEE.

Anoruo, M., et al. (2025). *Conference on Robot Learning (CoRL)*. (Under Review).

Conferences / Presentations

Apple Service Engineering Intern Spotlight	Summer 2023
MSRP Summer Research Symposium	Summer 2022
NSA On-Ramp Mission Research Symposium	Spring 2022
Undergraduate Research and Creative Achievement Day (URCAD)	Spring 2022
University of Pennsylvania SUNFEST Symposium	Summer 2021

Projects

Gesture Recognizing Drone

12/2023 – Present

- Developing machine learning model to recognize hand gestures signaling to land, takeoff, or follow a specific person.
- Developing Quadcopter drone to move around autonomously using a camera.

Interactive 3D Scene Reconstruction

9/2023 – 12/2023

- Adapted PyTorch's ResNet and MiDaS pretrained model to detect objects and their 3D position from a 2D image.
- Developed robust web scraping algorithm to query and download open-sourced 3D models from SketchFab.
- Utilized Panda3D engine to reconstruct 2D picture into 3D low fidelity simulation environment.

Song Genre Prediction

9/2023 – 12/2023

- Curated custom dataset to include song lyrics and genre using Spotify and Genius Lyric API.
- Developed Recurrent Neural Network model in PyTorch to predict song genre based on lyrics of the song.

Roblox Game Development

9/2021 – Present

- Served as the lead developer for fast paced action game accumulating close to 1 million players.
- Created analytical tools to track asset engagement and popularity.

Leadership / Extracurricular

Meyerhoff Peer Advisor	Fall 2022 – Present
Church Youth Leader	Fall 2021 – Present
National Society of Black Engineers (Executive Board)	Fall 2021 – Present
Creative Coders (Site Lead)	Spring 2023
Church Media Department	Summer 2018 – Spring 2023